

6.3 Verify and Troubleshoot EtherChannel

Verify and Troubleshoot EtherChannel

Verify EtherChannel

As always, when you configure devices in your network, you must verify your configuration. If there are problems, you will also need to be able to troubleshoot and fix them. There are a number of commands to verify an EtherChannel configuration:

- The **show interfaces port-channel** command displays the general status of the port channel interface.
- The **show etherchannel summary** command displays one line of information per port channel.
- The **show etherchannel port-channel** command displays information about a specific port channel interface.
- The **show interfaces etherchannel** command can provide information about the role of a physical member interface of the EtherChannel.

Common Issues with EtherChannel Configurations

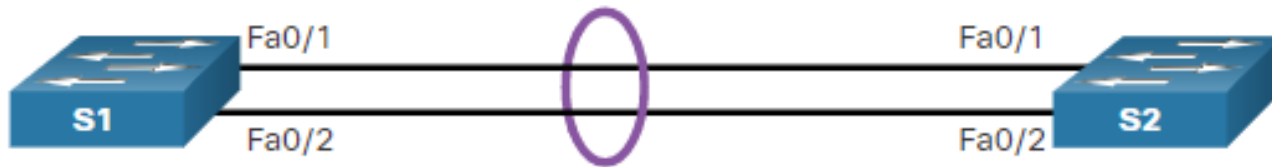
All interfaces within an EtherChannel must have the same configuration of speed and duplex mode, native and allowed VLANs on trunks, and access VLAN on access ports. Ensuring these configurations will significantly reduce network problems related to EtherChannel. Common EtherChannel issues include the following:

- Assigned ports in the EtherChannel are not part of the same VLAN, or not configured as trunks. Ports with different native VLANs cannot form an EtherChannel.
- Trunking was configured on some of the ports that make up the EtherChannel, but not all of them. It is not recommended that you configure trunking mode on individual ports that make up the EtherChannel. When configuring a trunk on an EtherChannel, verify the trunking mode on the EtherChannel.
- If the allowed range of VLANs is not the same, the ports do not form an EtherChannel even when PAgP is set to the **auto** or **desirable** mode.
- The dynamic negotiation options for PAgP and LACP are not compatibly configured on both ends of the EtherChannel.

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Troubleshoot EtherChannel Example

In the figure, interfaces Fa0/1 and Fa0/2 on switches S1 and S2 are connected with an EtherChannel. However, the EtherChannel is not operational.



Troubleshoot EtherChannel Example (Cont.)

Step 1. View the EtherChannel Summary Information: The output of the **show etherchannel summary** command indicates that the EtherChannel is down.

```
S1# show etherchannel summary
Flags: D - down          P - bundled in port-channel
      I - stand-alone s - suspended
      H - Hot-standby (LACP only)
      R - Layer3          S - Layer2
      U - in use          N - not in use, no aggregation
      f - failed to allocate aggregator
      M - not in use, minimum links not met
      m - not in use, port not aggregated due to minimum links not met
      u - unsuitable for bundling
      w - waiting to be aggregated
      d - default port
      A - formed by Auto LAG
Number of channel-groups in use: 1
Number of aggregators:          1
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SD)        -           Fa0/1(D)  Fa0/2(D)
```

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Troubleshoot EtherChannel Example (Cont.)

Step 2. View Port Channel Configuration: In the **show run | begin interface port-channel** output, more detailed output indicates that there are incompatible PAgP modes configured on S1 and S2.

```
S1# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!=====
S2# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
```

Troubleshoot EtherChannel Example (Cont.)

Step 3: Correct the Misconfiguration: To correct the issue, the PAgP mode on the EtherChannel is changed to desirable.

```
S1(config)# no interface port-channel 1
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
S1(config-if-range)# no shutdown
S1(config-if-range)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1#
```

Note: Port-Channel 1 removed and then re-added with the **channel-group** command, as opposed to directly changed. If one tries to change the configuration directly, STP errors cause the associated ports to go into blocking or errdisabled state

Troubleshoot EtherChannel Example (Cont.)

Step 4. Verify EtherChannel is Operational: The EtherChannel is now active as verified by the output of the **show etherchannel summary** command.

```
S1# show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:           1
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        PAgP        Fa0/1(P)  Fa0/2(P)
```